Moller scattering 4

From the first section we have

$$\sum_{abcd} \mathcal{M}_1 \mathcal{M}_1^* = \frac{e^4}{t^2} \left(8s^2 + 8u^2 - 64m^2 s - 64m^2 u + 192m^4\right)$$
$$\sum_{abcd} \mathcal{M}_1 \mathcal{M}_2^* = \frac{e^4}{tu} \left(-8s^2 + 64m^2 s - 96m^4\right)$$
$$\sum_{abcd} \mathcal{M}_2 \mathcal{M}_2^* = \frac{e^4}{u^2} \left(8s^2 + 8t^2 - 64m^2 s - 64m^2 t + 192m^4\right)$$

We also have by computer algebra

$$\sum_{abcd} \mathcal{M}_1 \mathcal{M}_1^* = \frac{e^4}{t^2} \left(32(p_1 \cdot p_2)^2 + 32(p_1 \cdot p_4)^2 - 64m^2(p_1 \cdot p_2) + 64m^2(p_1 \cdot p_4)\right)$$
$$\sum_{abcd} \mathcal{M}_1 \mathcal{M}_2^* = \frac{e^4}{tu} \left(-32(p_1 \cdot p_2)^2 + 64m^2(p_1 \cdot p_2)\right)$$
$$\sum_{abcd} \mathcal{M}_2 \mathcal{M}_2^* = \frac{e^4}{u^2} \left(32(p_1 \cdot p_2)^2 + 32(p_1 \cdot p_3)^2 - 64m^2(p_1 \cdot p_2) + 64m^2(p_1 \cdot p_3)\right)$$

For $m = 0$

$$32(p_1 \cdot p_2)^2 = 8s^2$$
$$32(p_1 \cdot p_3)^2 = 8t^2$$
$$32(p_1 \cdot p_4)^2 = 8u^2$$

Eigenmath script